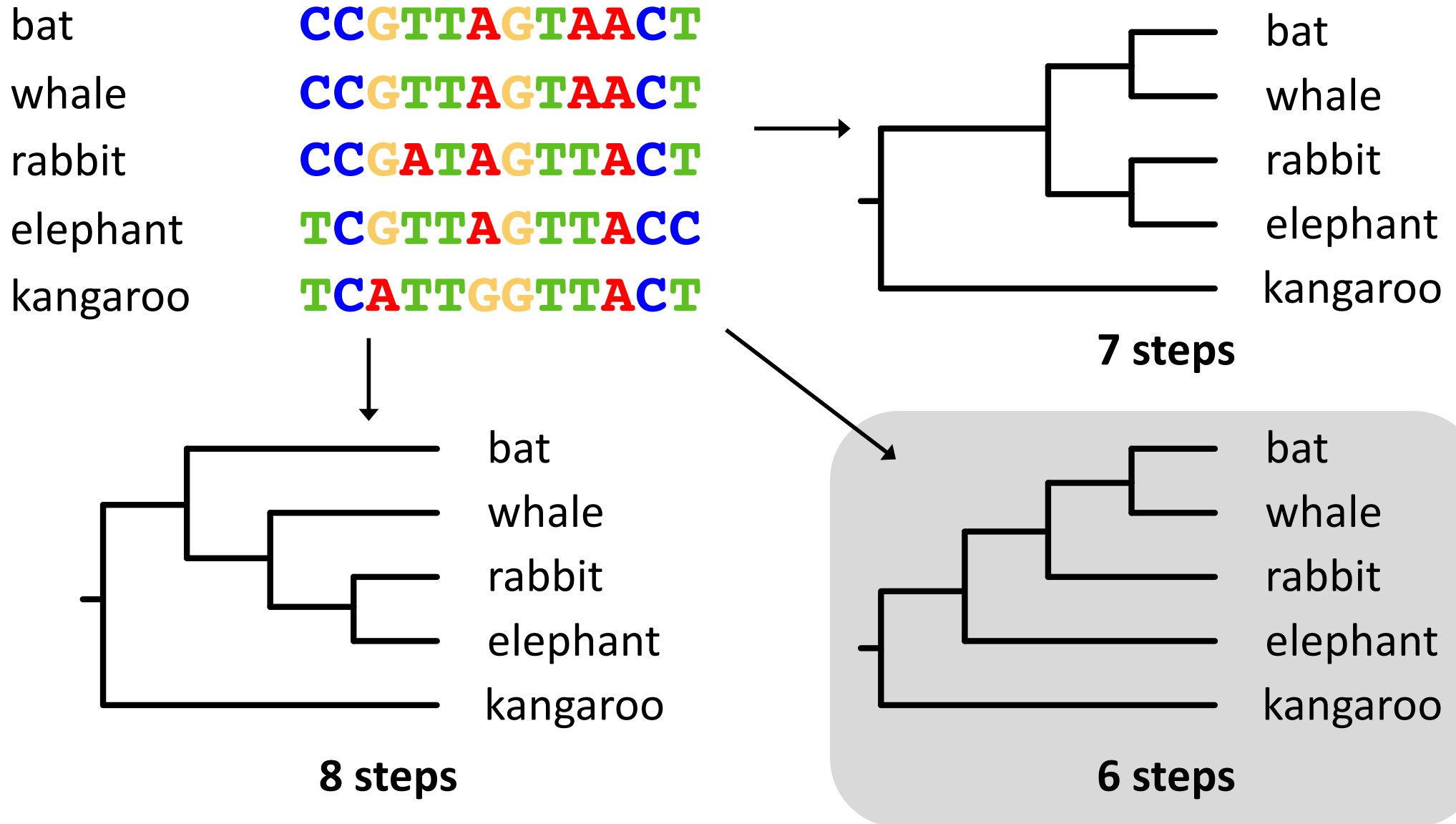

Lecture 1.4

Phylogenetic Methods

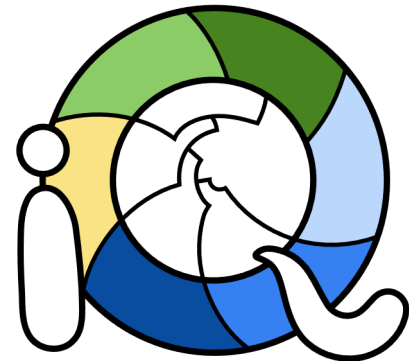
Maximum parsimony



Popular phylogenetic methods

1. Maximum parsimony
2. Distance-based methods
3. Maximum likelihood
4. Bayesian inference

Model-based methods



Distance-Based Methods

Neighbour joining

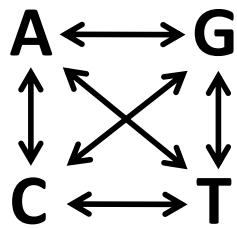
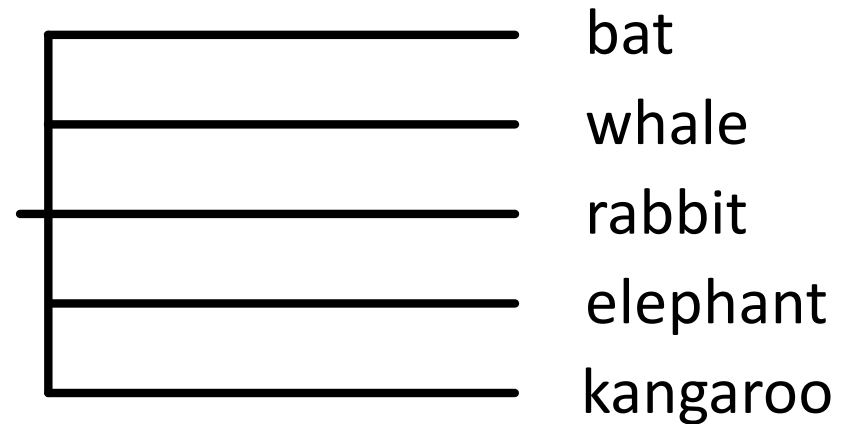
bat **CCGTTAGTA**ACT

whale **CCGTTAGTA**ACT

rabbit **CCG****A****T**AGTTACT

elephant **T****C****G****T****T****A****G****T****T****A****C****C**

kangaroo **T****C****A****T****T****G****G****T****T****A****C****T**

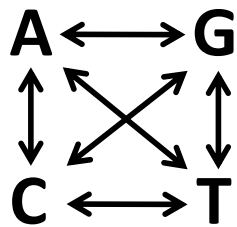
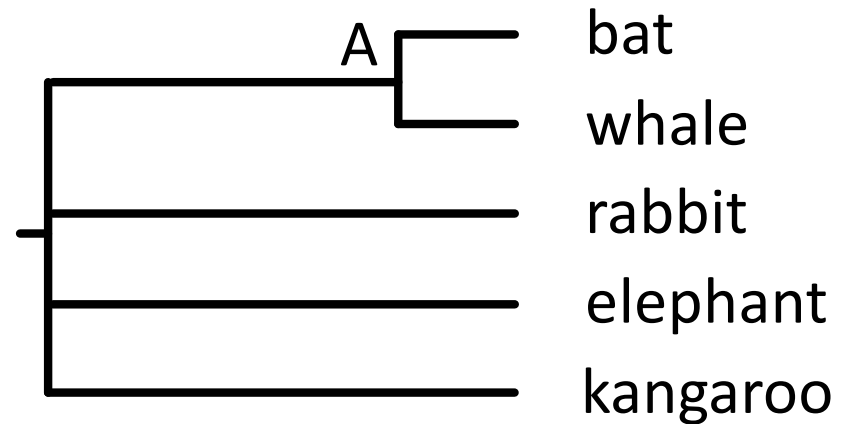


	bat	whale	rabbit	elephant	kangaroo
bat	–				
whale	.15	–			
rabbit	.20	.25	–		
elephant	.35	.40	.35	–	
kangaroo	.55	.60	.55	.55	–

**Clustering
algorithm**

Neighbour joining

bat **CCGTTAGTAACT**
 whale **CCGTTAGTAACT**
 rabbit **CCGATAGTTACT**
 elephant **TCGTTAGTTACC**
 kangaroo **TCATTGGTTACT**

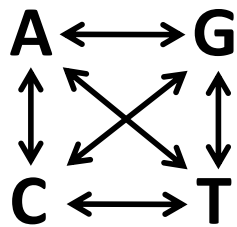
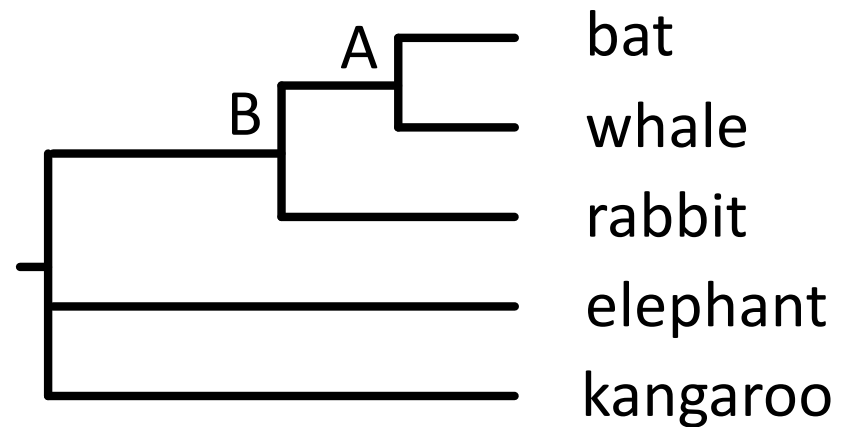


	bat	whale	rabbit	elephant	kangaroo
bat	-				
whale	.15	-			
rabbit	.20	.25	-		
elephant	.35	.40	.35	-	
kangaroo	.55	.60	.55	.55	-

**Clustering
algorithm**

Neighbour joining

bat	CCGTTAGTAACT
whale	CCGTTAGTAACT
rabbit	CCGATAGTTACT
elephant	TCGTTAGTTACC
kangaroo	TCATTGGTTACT

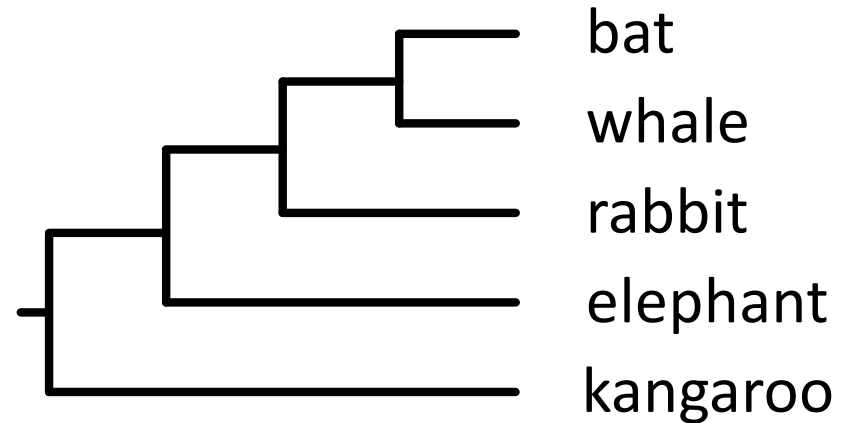


	A	rabbit	elephant	kangaroo
A	-			
rabbit	.15	-		
elephant	.30	.35	-	
kangaroo	.50	.55	.60	-

**Clustering
algorithm**

Neighbour joining

bat	CCGTTAGTAACT
whale	CCGTTAGTAACT
rabbit	CCGATAGTTACT
elephant	TCGTTAGTTACC
kangaroo	TCATTGGTTACT



Distance-based methods

- **Clustering algorithms**
 - Unweighted pair group method with arithmetic mean (UPGMA)
 - Neighbour joining
- **Tree searching using optimality criteria**
 - Minimum evolution
 - Least-squares inference

Strengths and weaknesses

- **Strengths**

- Very quick
- Deals with multiple substitutions and long-branch attraction

- **Weaknesses**

- Loss of information in pairwise comparisons
- Unable to implement sophisticated evolutionary models

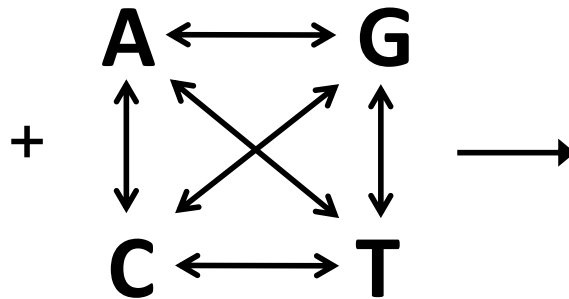
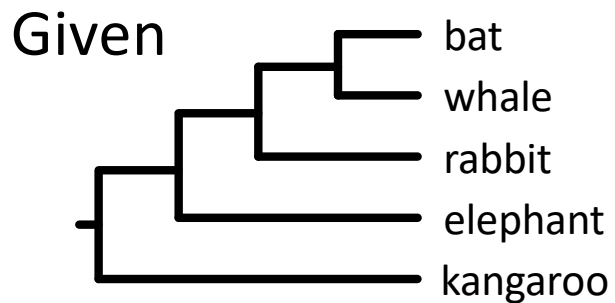
Maximum Likelihood

Maximum likelihood

Likelihood of hypothesis $H =$

$$P(D|H)$$

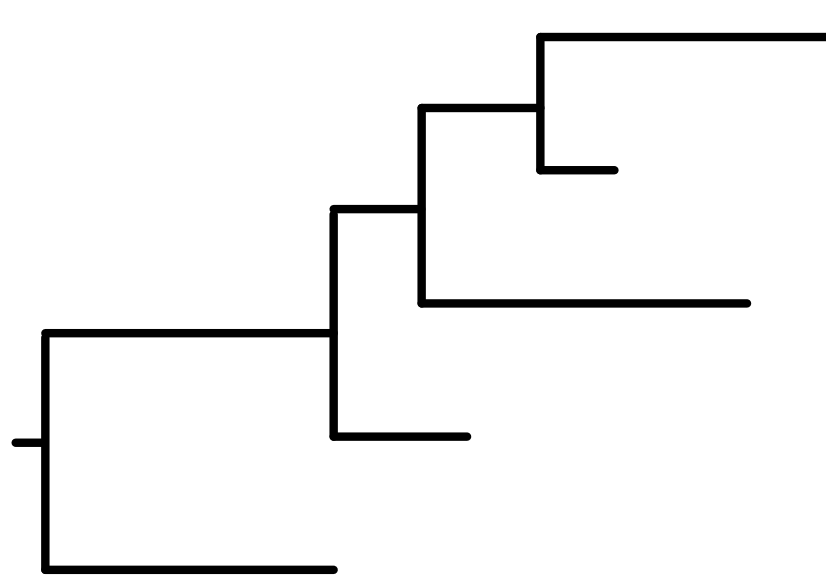
Probability of the data, given the hypothesis



Probability of?

bat	CCGTTAGTAACT
whale	CCGTTAGTAACT
rabbit	CCGATAGTTACT
elephant	TCGTTAGTTACC
kangaroo	TCATTGGTTACT

Maximum likelihood



bat

CCGTTAGTAACT****

whale

CCGTTAGTAACT****

rabbit

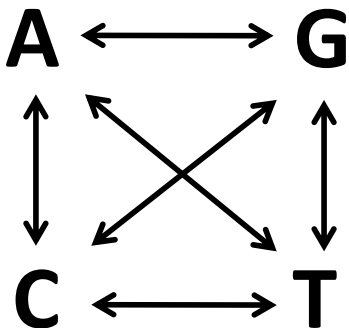
CCGATAGTTACT

elephant

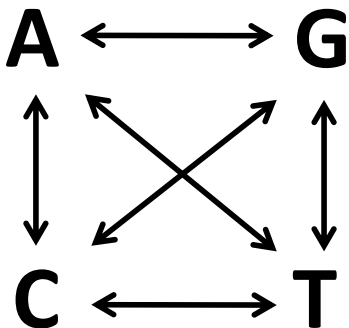
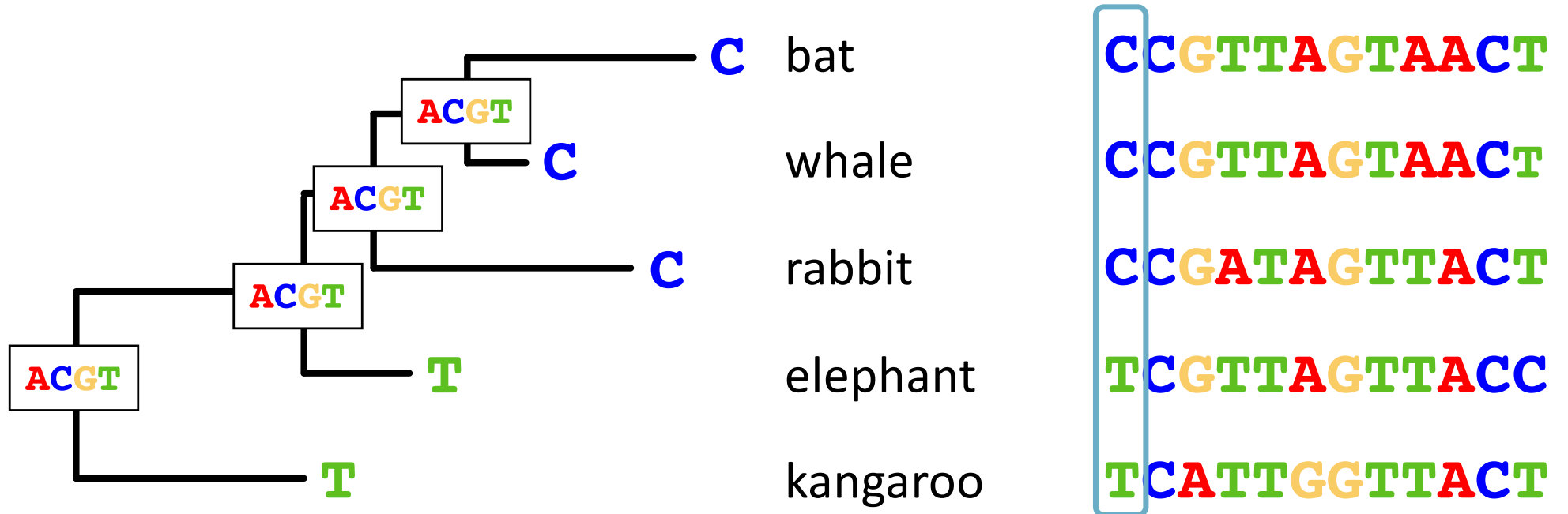
TCGTTAGTTACC

kangaroo

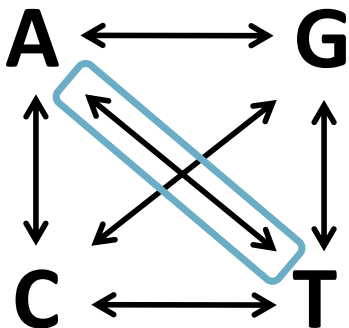
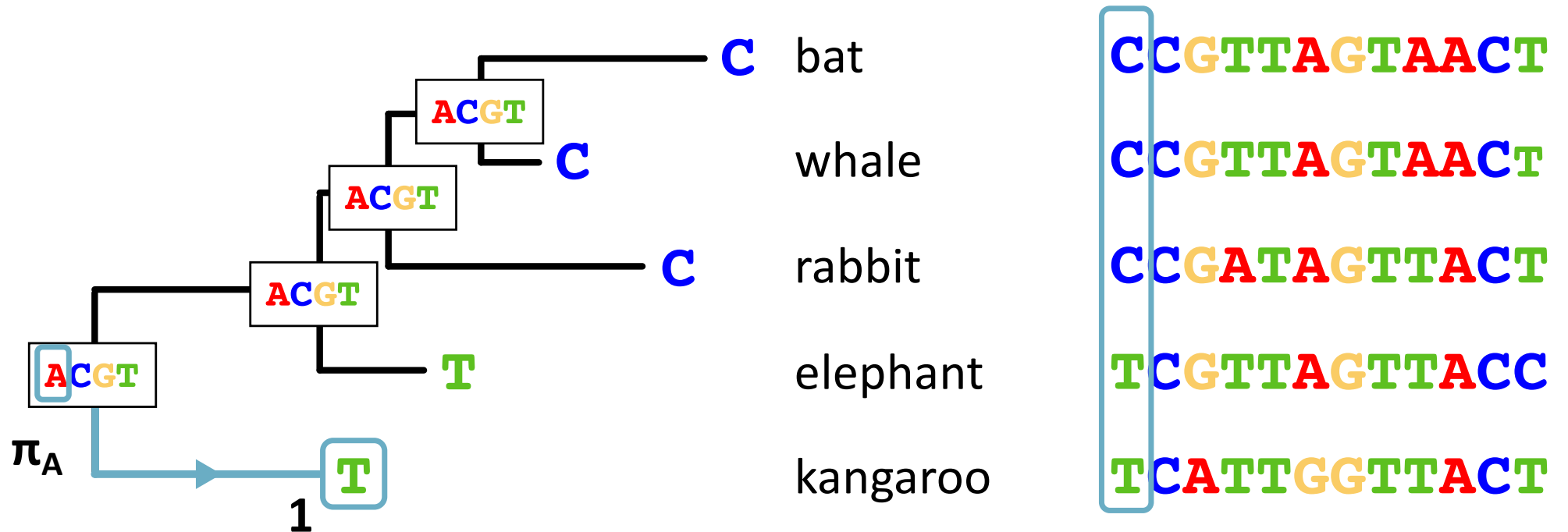
TCATTGGTTACT



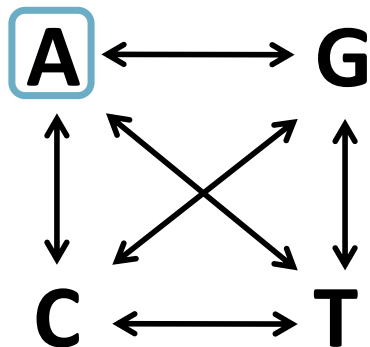
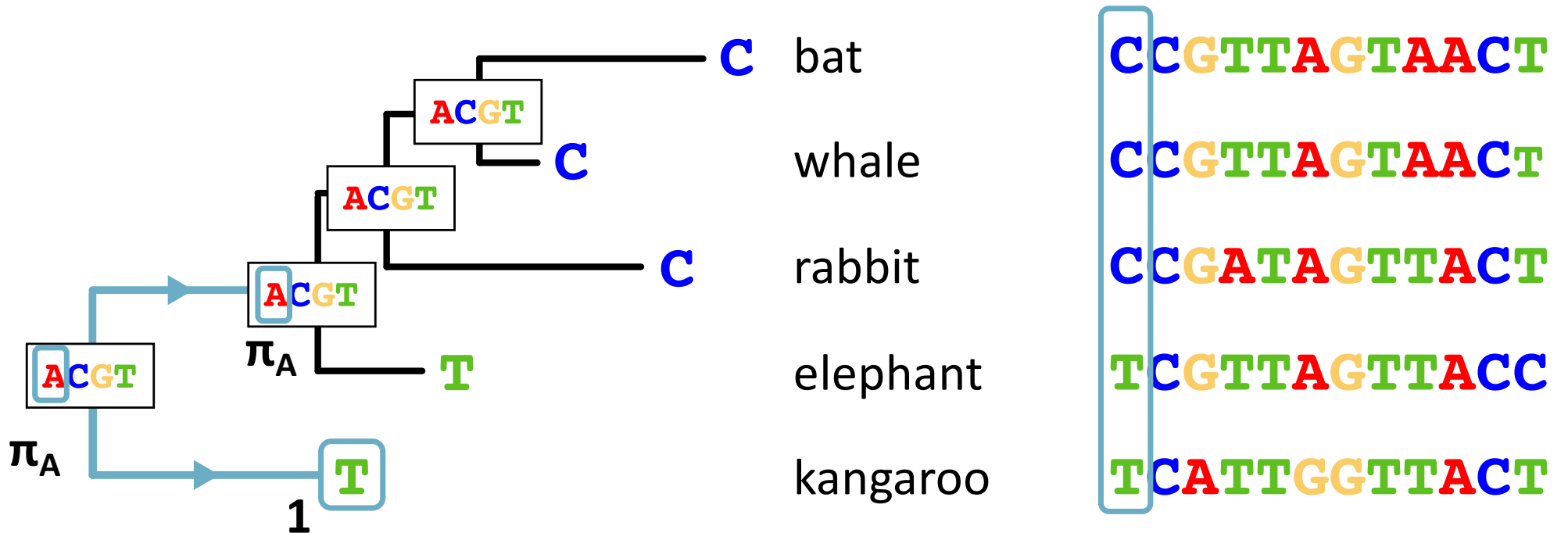
Maximum likelihood



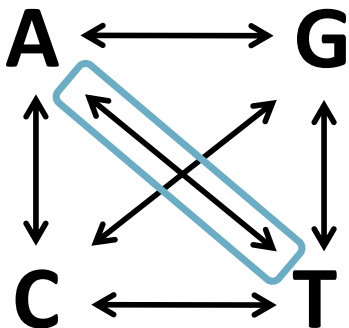
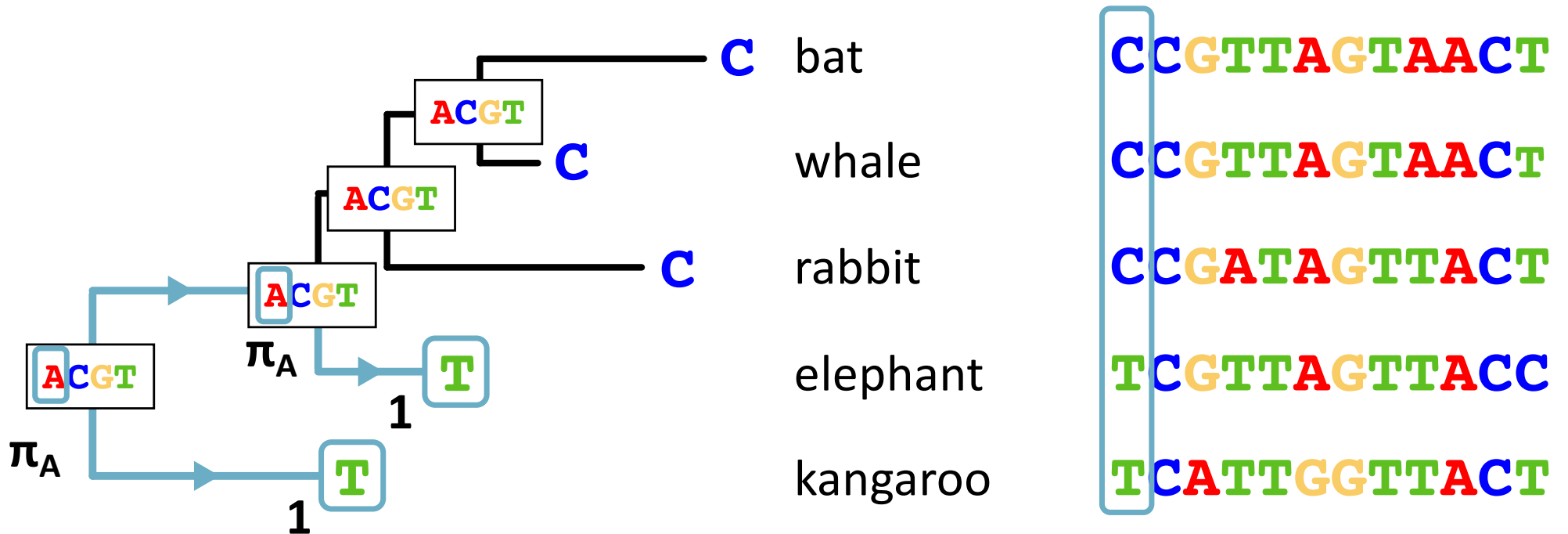
Maximum likelihood



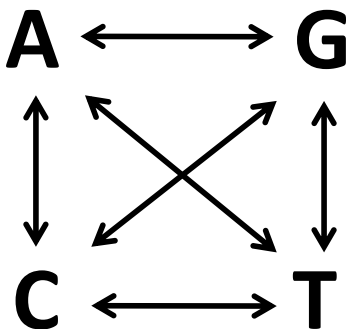
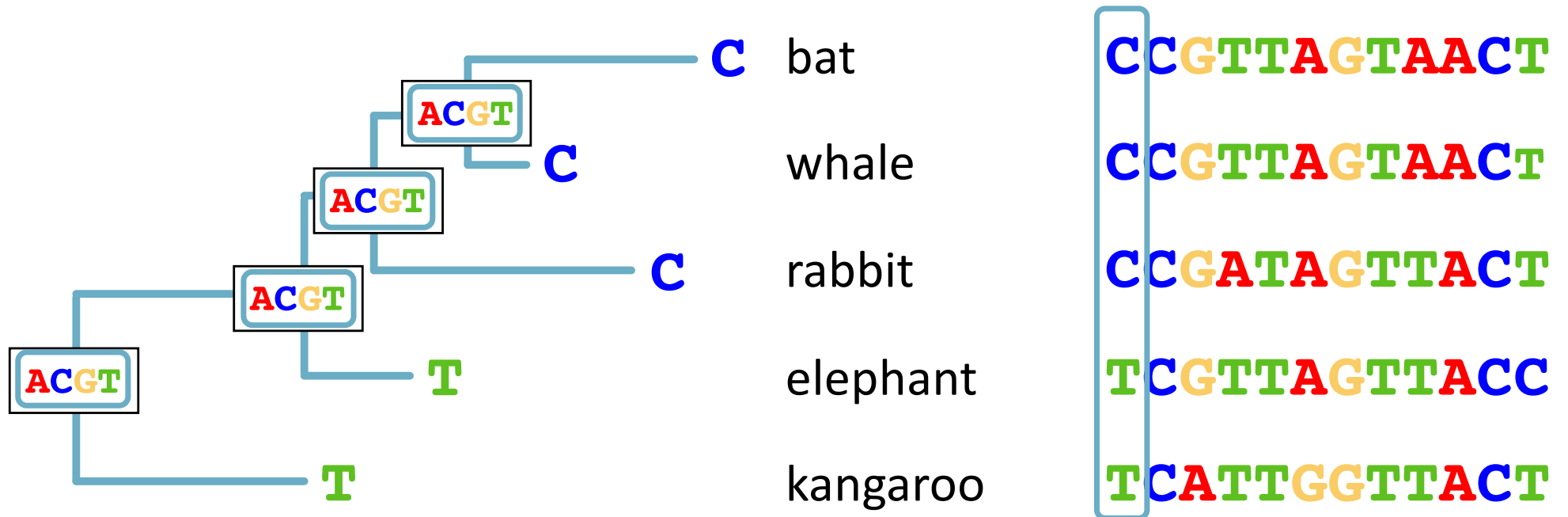
Maximum likelihood



Maximum likelihood

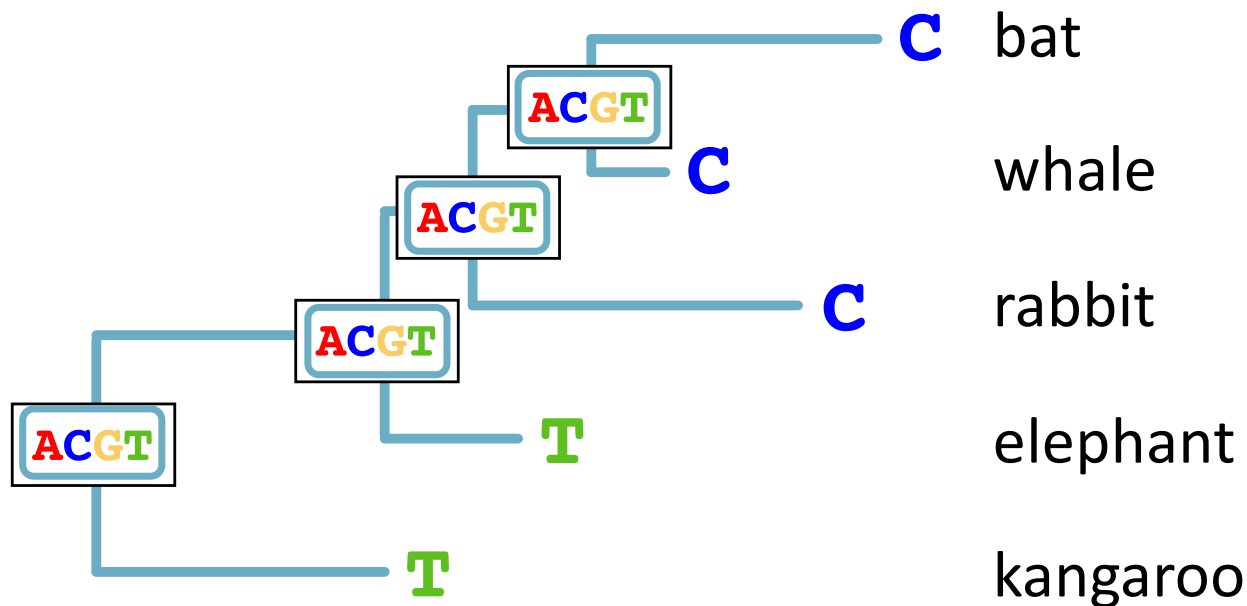


Maximum likelihood

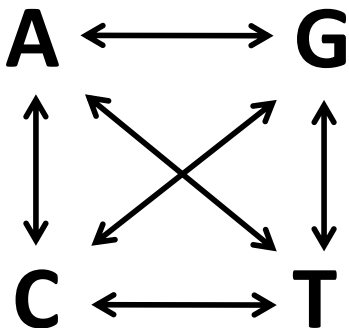


Likelihood is summed over all possibilities

Maximum likelihood



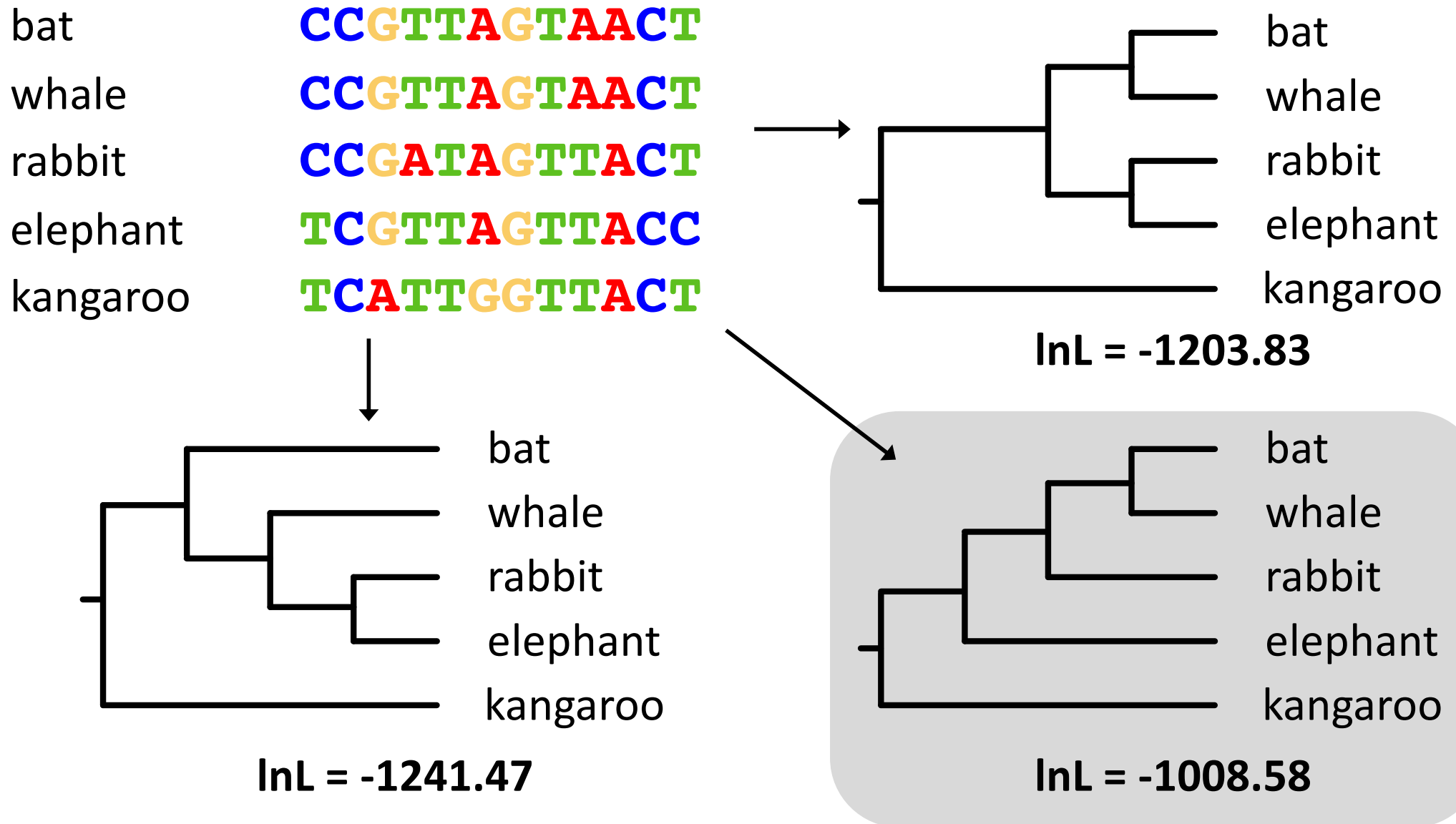
CCGTTAGTAACT
CCGTTAGTAACT
CCGATAGTTACT
TCGTTAGTTACC
TCATTGGTTACT



Likelihood is multiplied across all sites

Very low probability of observing
any particular alignment

Maximum likelihood



Likelihood optimisation

- Search through the space of possible trees (including branch lengths) and model parameter values
- Calculate the likelihood for these
- Find best tree and model parameter values
- Multivariate optimisation

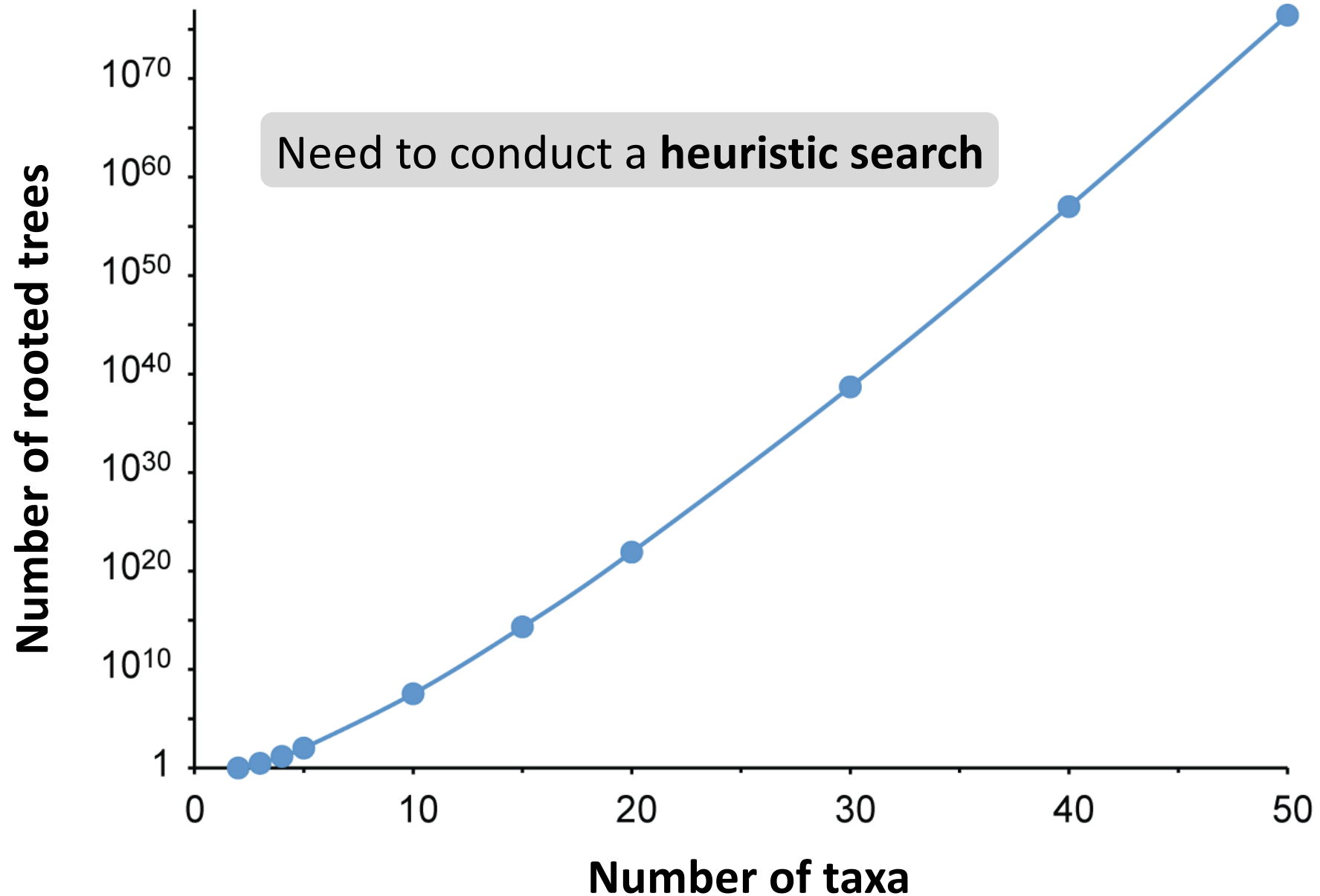
Finding the best tree

- For n taxa, the number of possible unrooted trees (B_n) is:

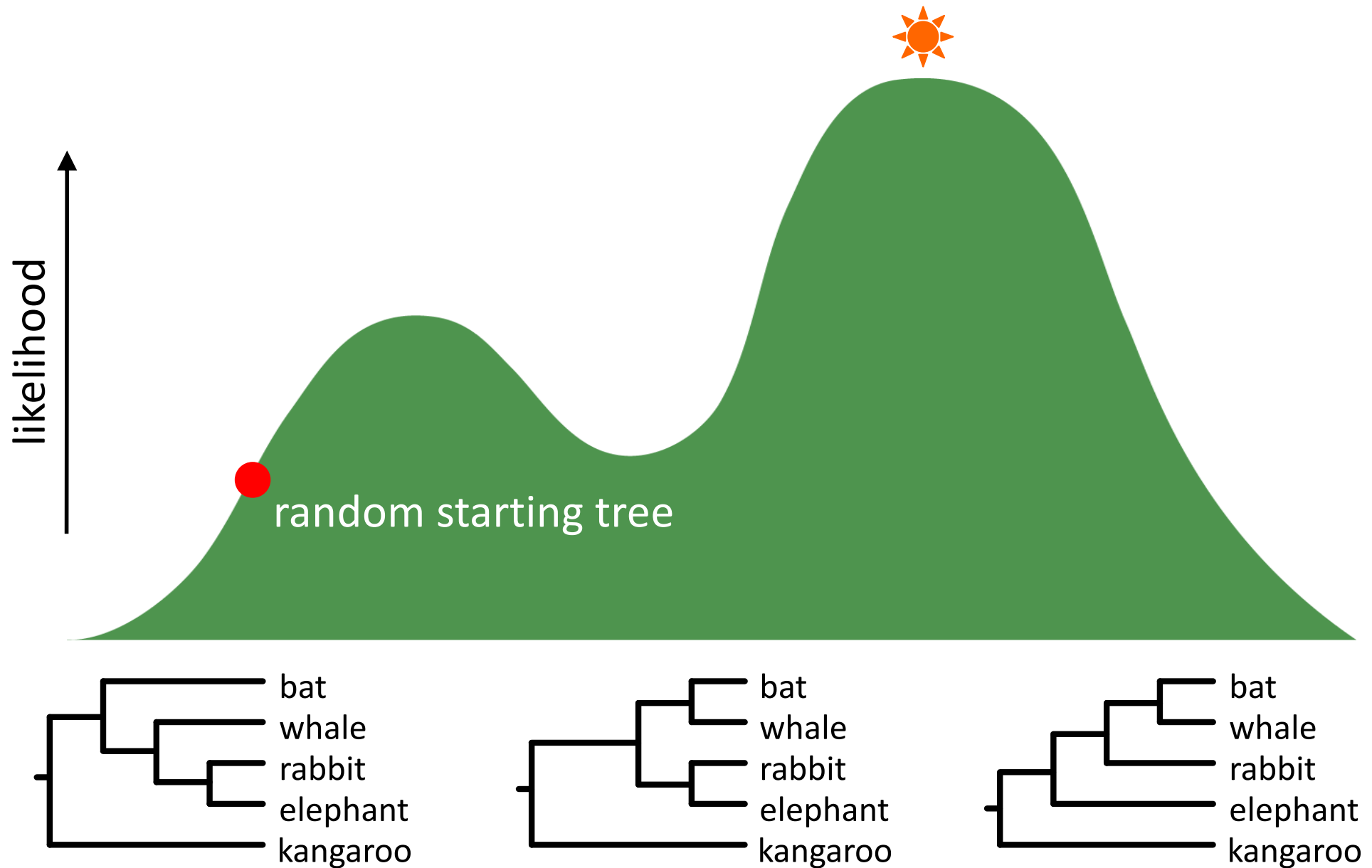
$$B_n = 1 \times 3 \times 5 \times \dots \times (2n - 5) = \prod_{i=3}^n (2i - 5)$$

- For example:
 - 4 taxa $\rightarrow$ 3 trees
 - 5 taxa $\rightarrow$ 15 trees
 - 10 taxa $\rightarrow$ 2,027,025 trees

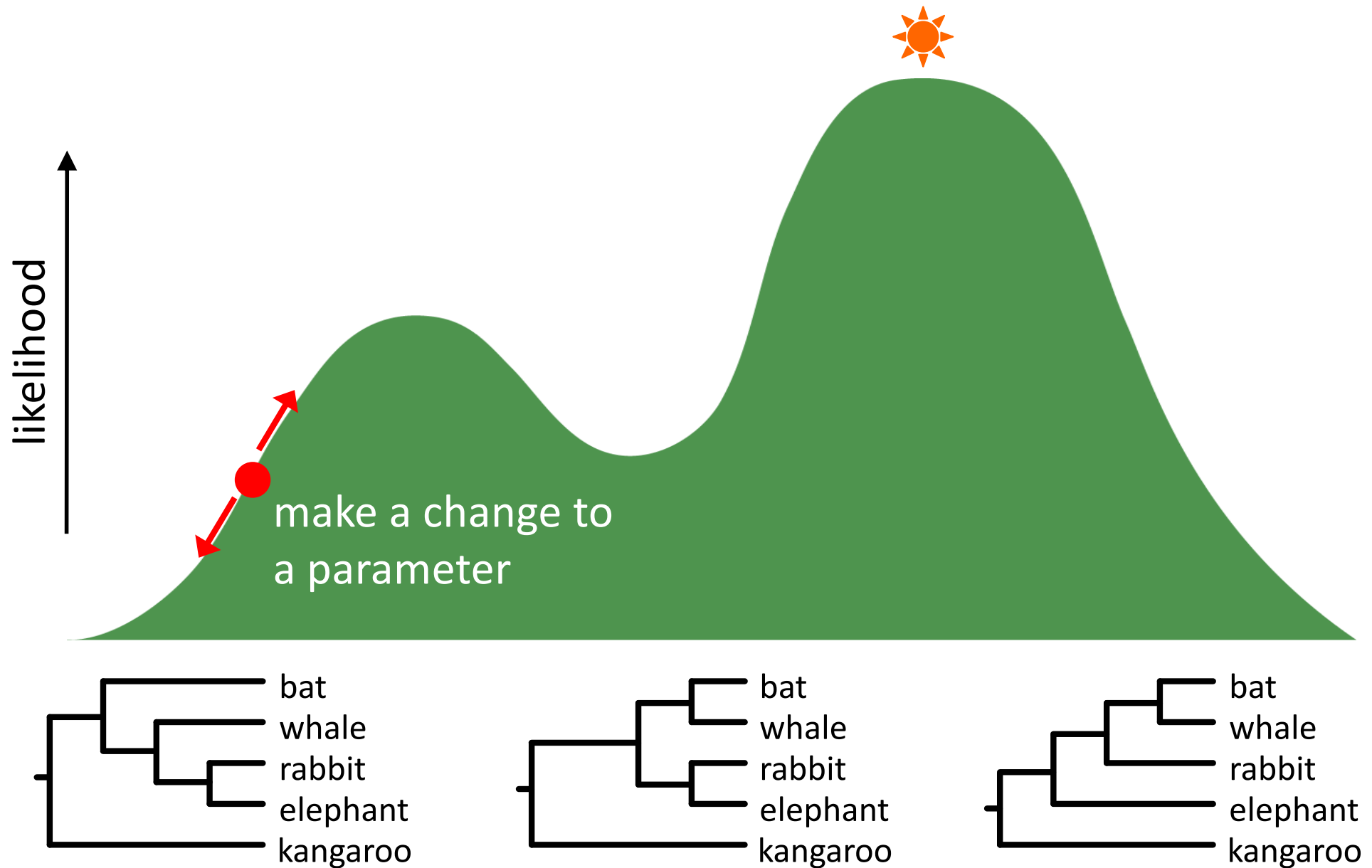
Finding the best tree



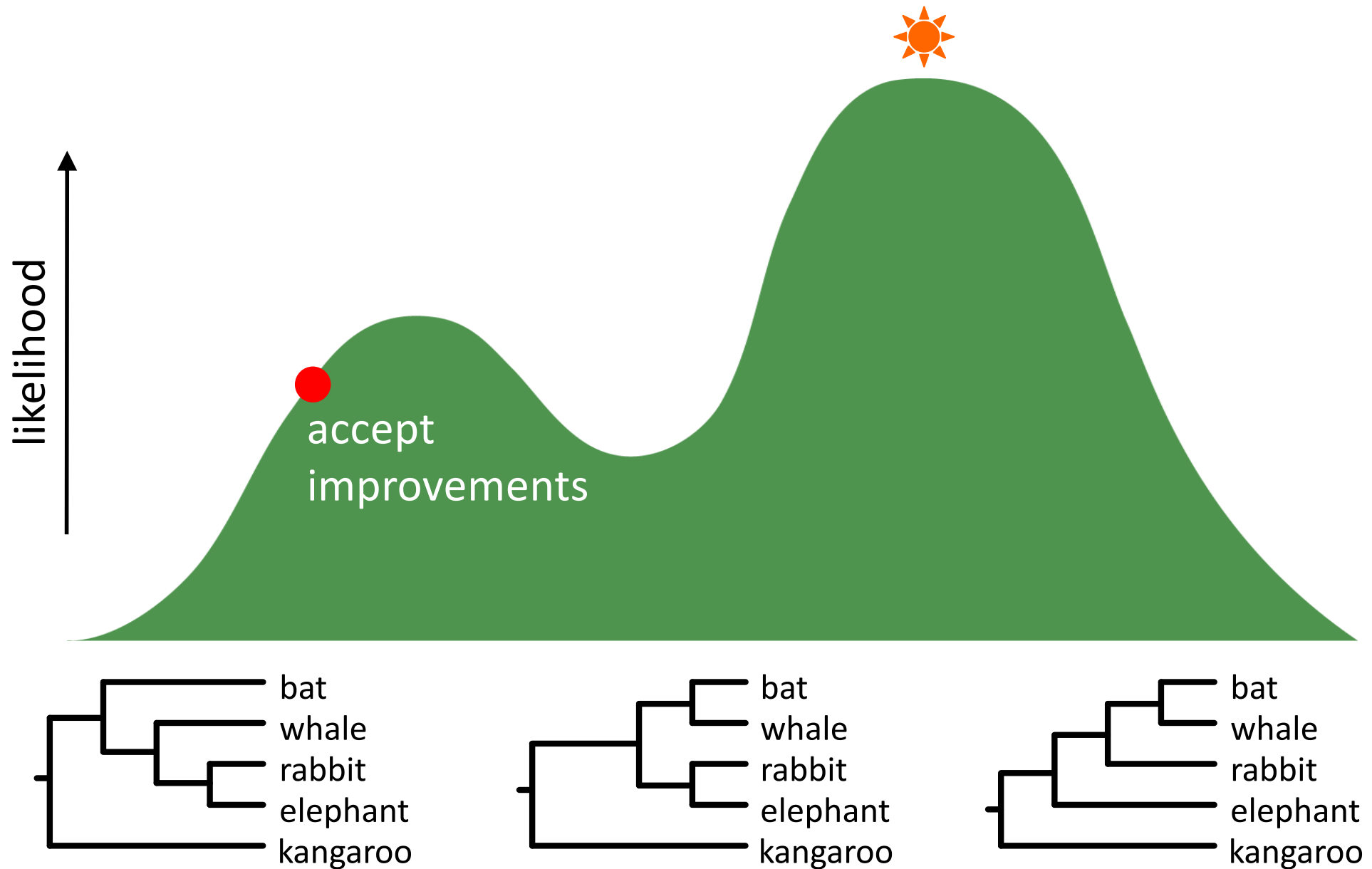
Heuristic search



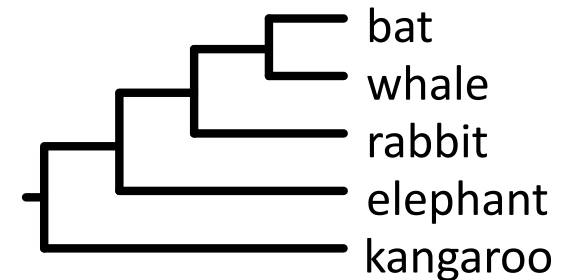
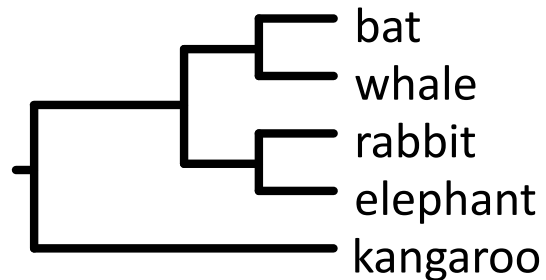
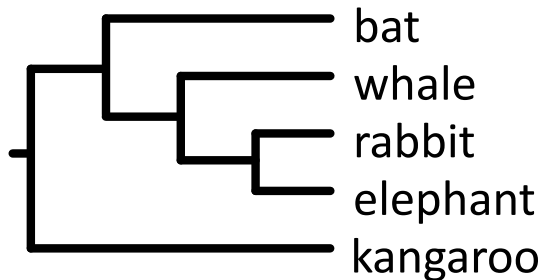
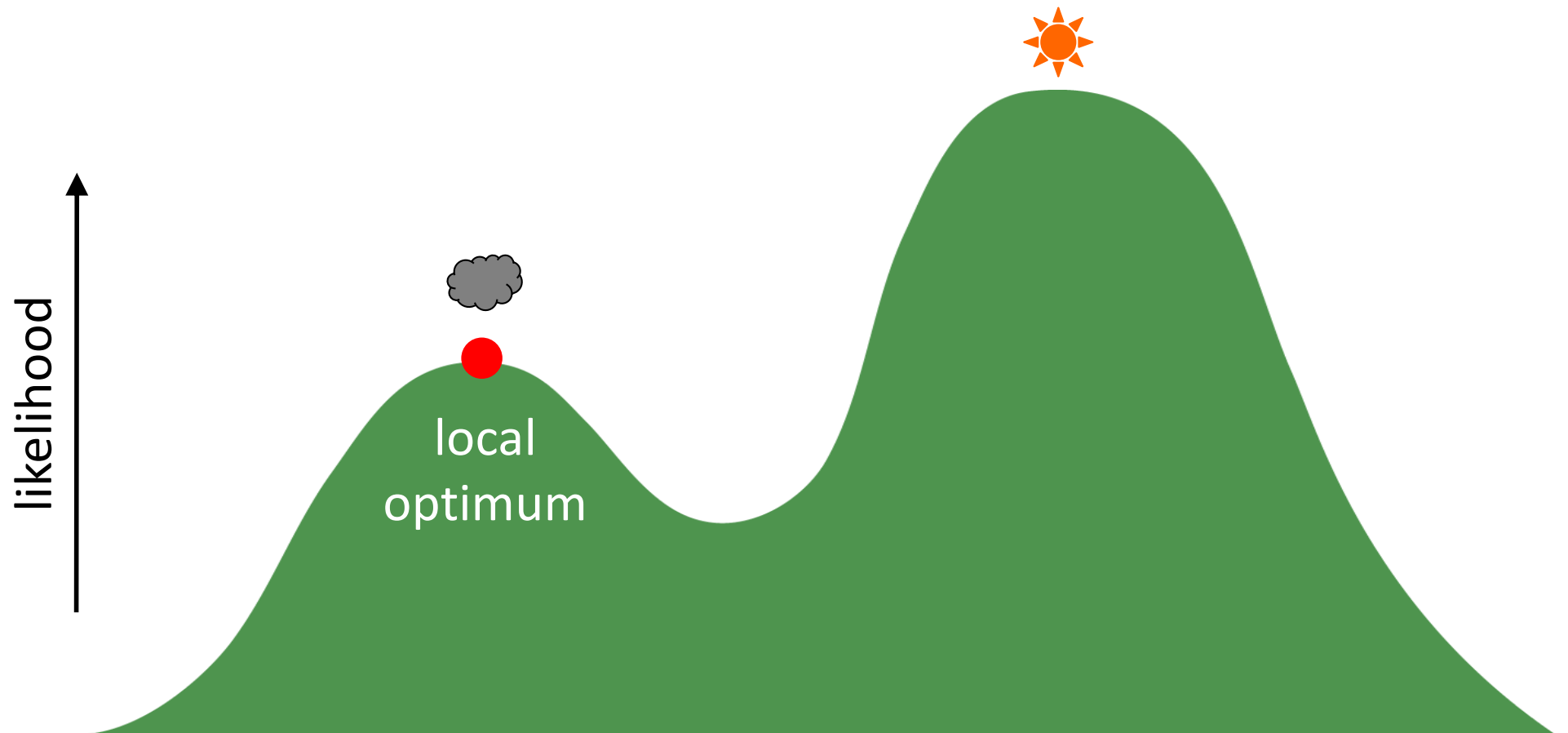
Heuristic search



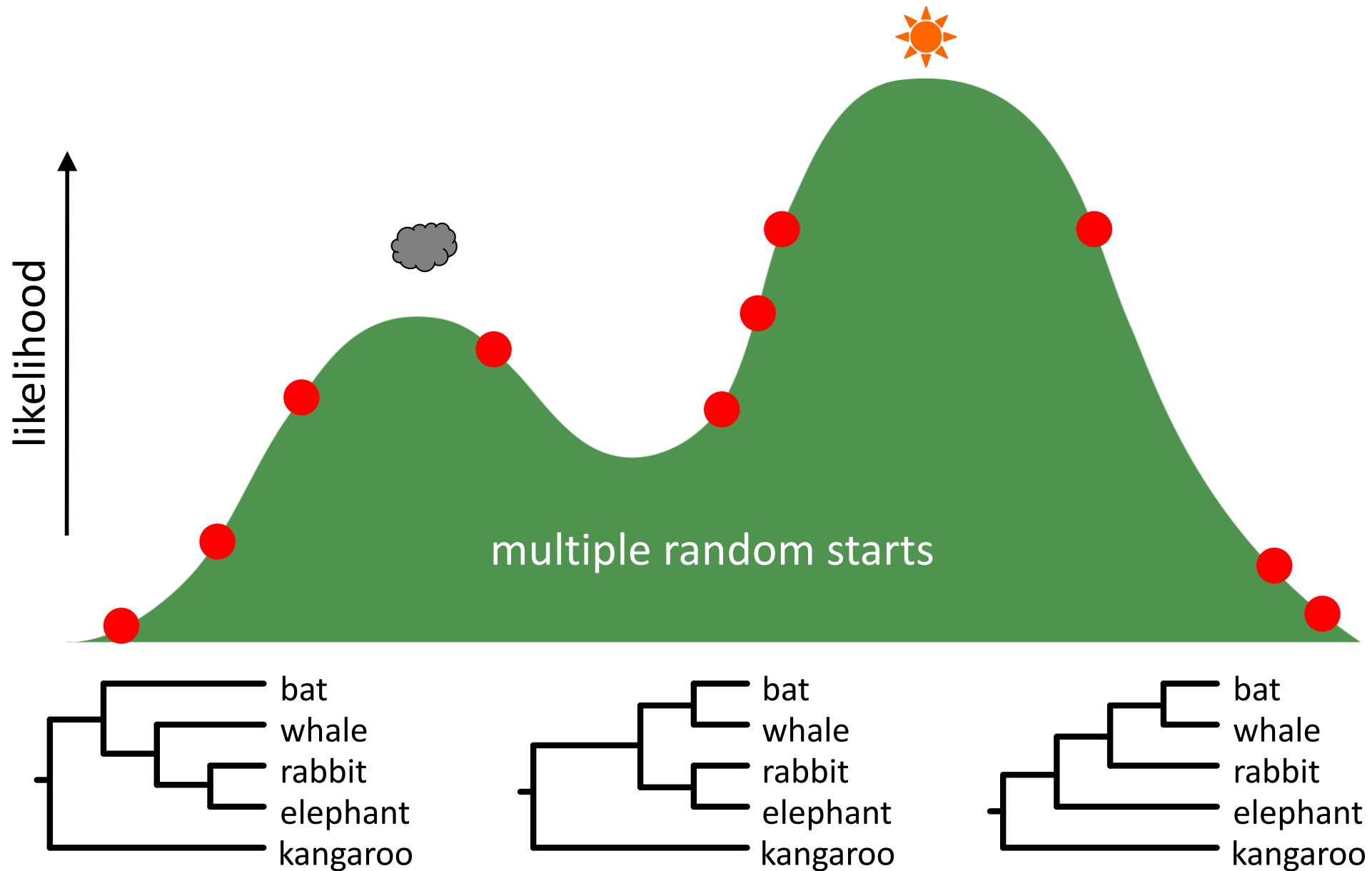
Heuristic search



Heuristic search

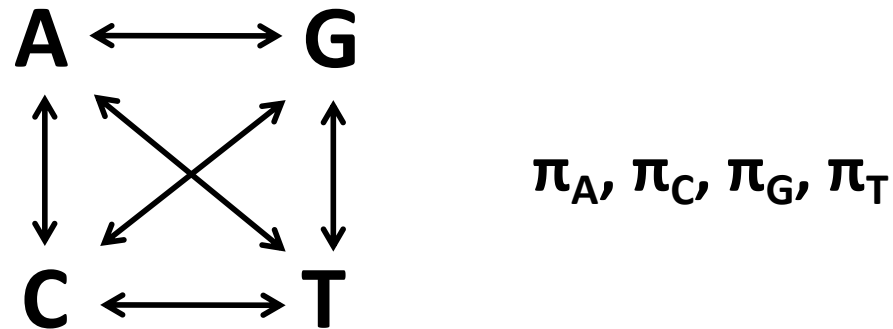


Heuristic search

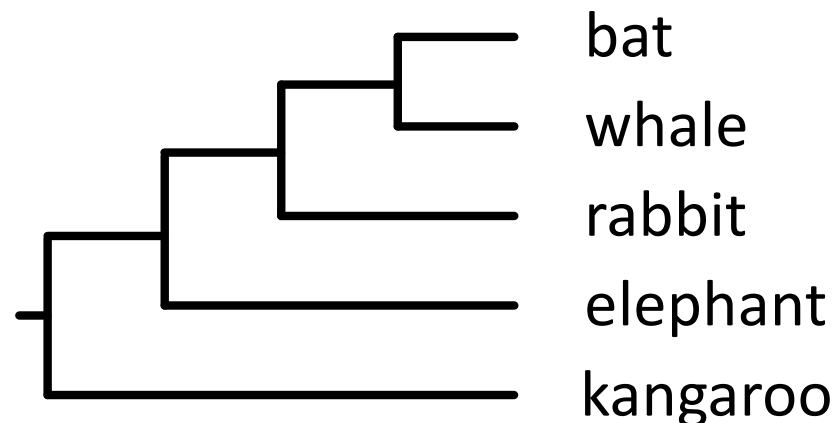


Maximum-likelihood estimates

A single set of maximum-likelihood estimates of model parameters



A single maximum-likelihood tree



Strengths and weaknesses

- **Strengths**

- Rigorous statistical method
- Deals with multiple substitutions and long-branch attraction
- Robust to violations of assumptions

- **Weaknesses**

- Generally not feasible to implement very parameter-rich models
- Searching tree space can be difficult

Software

RAxML



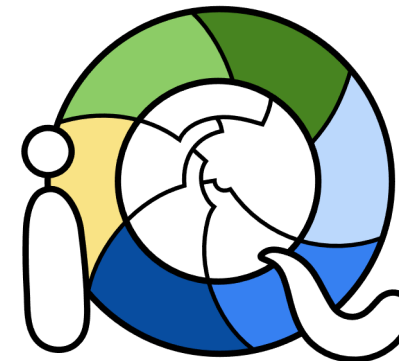
PhyML



MEGA



PAML



IQ-TREE

Bootstrapping

Nonparametric bootstrap

- Uncertainty in the estimate of the tree is inferred indirectly using **bootstrapping analysis**
- “pull oneself up by one's bootstraps”
- Bootstrapping analysis can be used in various phylogenetic methods:
 - Maximum parsimony
 - Distance-based methods
 - Maximum likelihood



Bootstrapping

bat	CCGTTAGTAACT
whale	CCGTTAGTAACT
rabbit	CCGATAGTTACT
elephant	TCGTTAGTTACC
kangaroo	TCATTGGTTACT

Randomly sample sites (with replacement)

bat	T
whale	T
rabbit	A
elephant	T
kangaroo	T

Bootstrapping

bat	CCGTTAGTAACT
whale	CCGTTAGTAACT
rabbit	CCGATAGTTACT
elephant	TCGTTAGTTACC
kangaroo	TCATTGGTTACT

bat	TG
whale	TG
rabbit	AG
elephant	TG
kangaroo	TG

Bootstrapping

bat	CCGTTAGTAACT
whale	CCGTTAGTAACT
rabbit	CCGATAGTTACT
elephant	TCGTTAGTTACC
kangaroo	TCATTGGTTACT

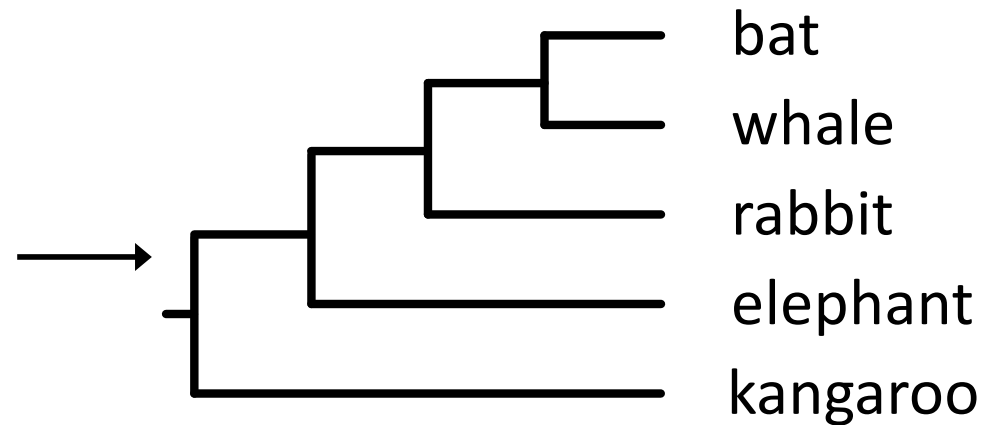
bat	TGCCCTTAGCAC
whale	TGCCCTTAGCAC
rabbit	AGCCCATAGCAC
elephant	TGCTCTCAGCAT
kangaroo	TGCTCTTAACGT

Bootstrapping

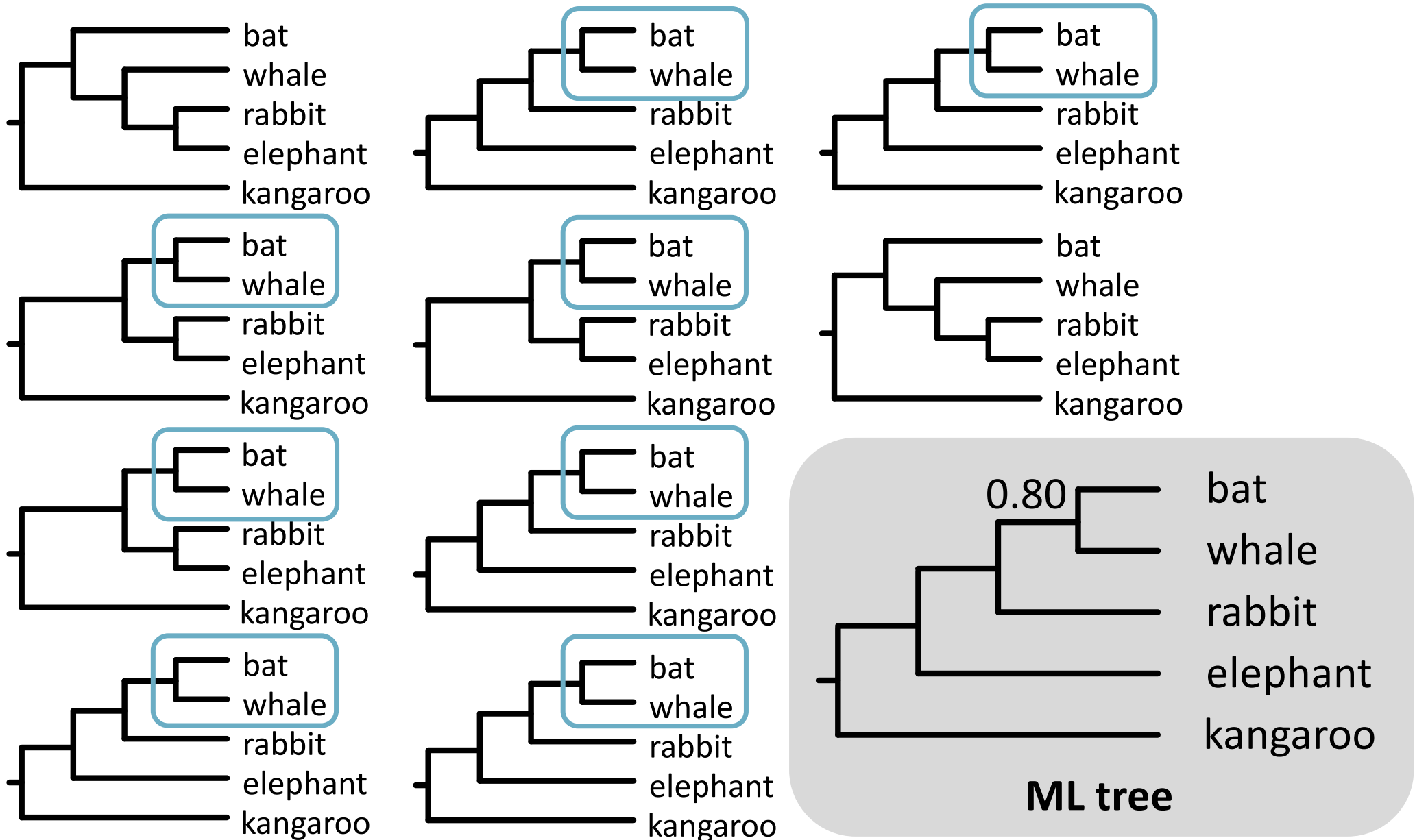
bat	CCGTTAGTAACT
whale	CCGTTAGTAACT
rabbit	CCGATAGTTACT
elephant	TCGTTAGTTACC
kangaroo	TCATTGGTTACT

Repeat 1,000 times

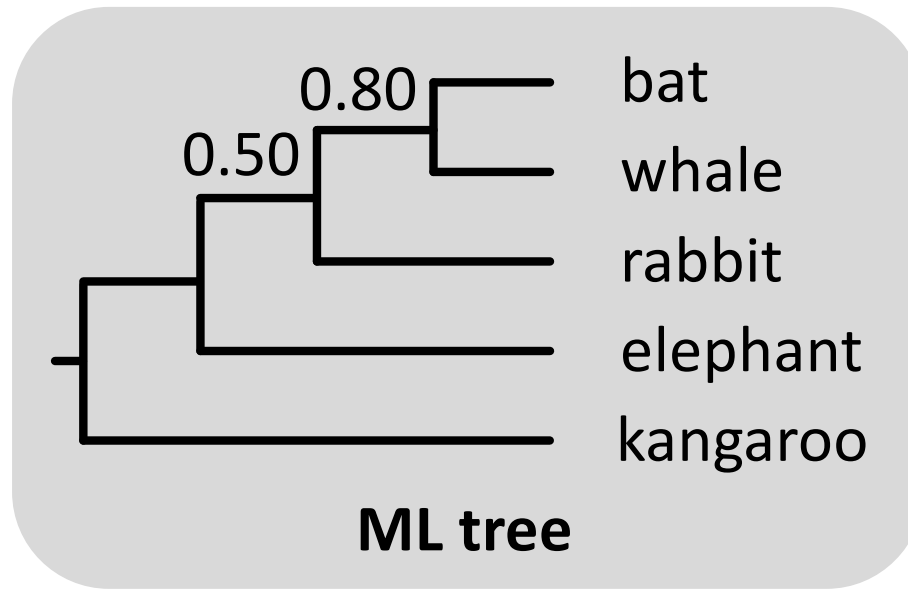
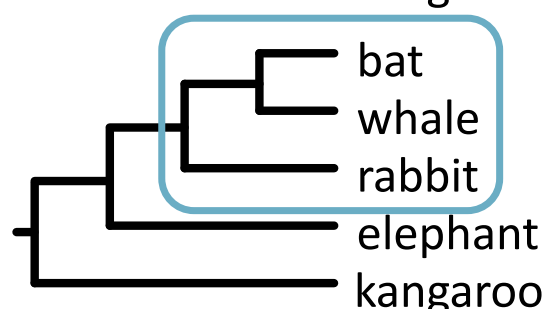
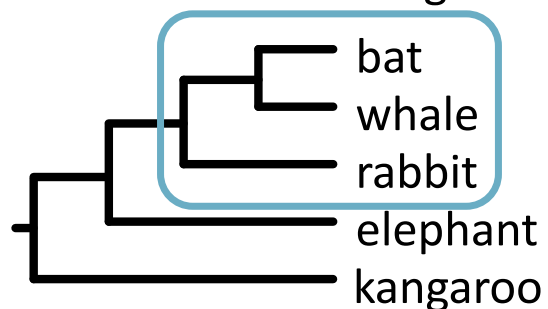
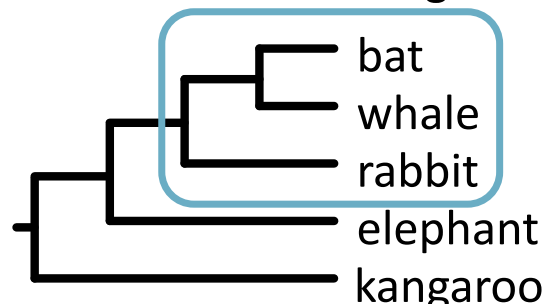
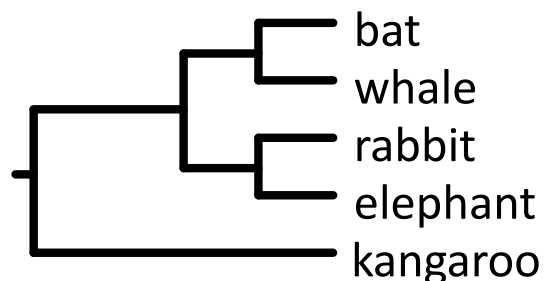
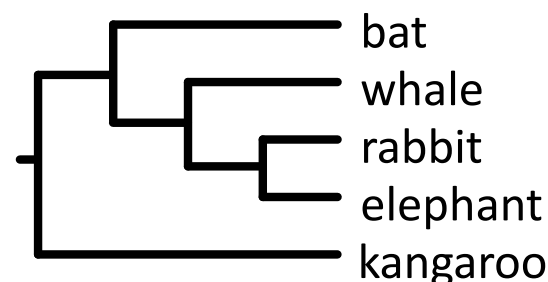
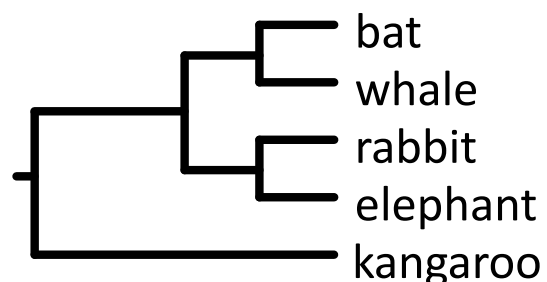
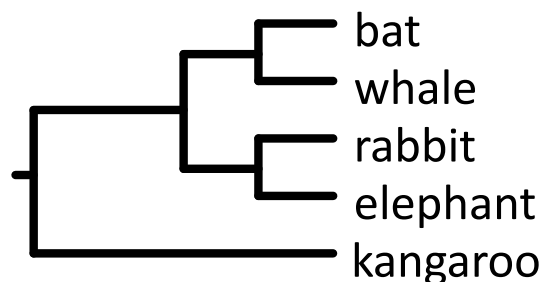
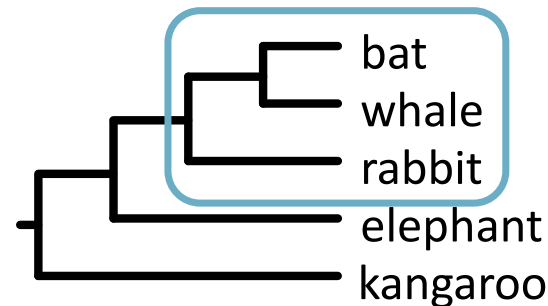
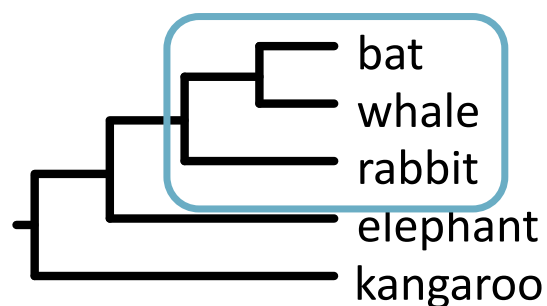
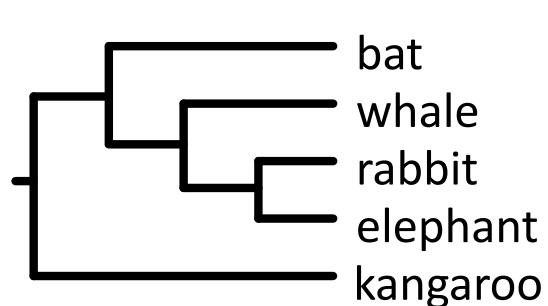
bat	TGCCCTTAGCAC
whale	TGCCCTTAGCAC
rabbit	AGCCCATAGCAC
elephant	TGCTCTCAGCAT
kangaroo	TGCTCTTAACGT



Bootstrapping



Bootstrapping



Interpreting bootstrap values

- **Felsenstein (1985)**

bootstrapping provides a confidence interval that contains the *phylogeny that would be estimated from repeated sampling of many characters from the underlying set of all characters*

- Bootstrap values are **measures of repeatability**

- Interpretation: 70% moderate support, 95% high support
- High when the data set is large
- Not meaningful when analysing genome-scale data

Methods in practice

- **Maximum parsimony**
 - Commonly used to analyse morphological data
 - Rarely used to analyse molecular data
- **Distance-based methods**
 - Popular in some fields of research
 - Used to analyse very large data sets with many taxa
- **Maximum likelihood**
 - Widely used, lost some ground to Bayesian methods but is experiencing a resurgence (thanks to rapid ML methods)

Useful references

- **Molecular phylogenetics: principles and practice**
Yang & Rannala (2012) *Nature Reviews Genetics* 13: 303–314.

